LAMP: ASP Rule Induction and Optimization with LLMs

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Headline

LAMP asks an LLM to recover ASP rules from *instance facts* and an *expected answer set*, then checks every candidate with Clingo. Across 120 synthetic rule-induction benchmarks, the strongest LAMP models nearly matched ILASP, solved every ILASP timeout case, and usually produced smaller, more stratified programs than the synthetic originals.

99.2%
Best final accuracy with LAMP
gpt-5-mini: 119 / 120

98.3%
Best local model result
gpt-oss:20b: 118 / 120

3 / 3
ILASP timeout benchmarks solved
by both strong LAMP models

3–6
Average output literals for correct
solutions vs. 22 or 33 originals

Problem Setting

Writing ASP that is both correct and efficient is hard. LAMP studies a focused version of that challenge: recover a rule set from example facts and the stable model it should produce.

$$AS(F \cup \Pi) = \{A^*\}$$

Here F is the given fact set, A^* is the target answer set, Φ is a bounded hypothesis space, and the goal is to find a small program $\Pi \subseteq \Phi$ that reproduces A^* exactly.

- Both LAMP and ILASP receive the same facts, target answer set, and rule bounds.
- This makes the comparison about *search strategy*: LLM-guided generation versus conflict-driven symbolic search.
- The task is intrinsically difficult: bounded-arity non-ground normal ASP reasoning is Σ_2^P-complete, and learning adds another search layer.

Toy Benchmark

Facts

```
d0(o0).
d0(o1).
d1(o1).
```

Target answer set

```
{ d0(o0), d0(o1), d1(o1), d2(o0) }
```

One minimal solution

```
d2(X) :- d0(X), not d1(X).
```

Correctness is judged by the stable model, not by reproducing the exact synthetic source rules. Many different rule sets can yield the same answer set.

Benchmark Design

Parameter	Values used
Predicates	6, 10
Terms	6, 10
Facts	5, 6, 9
Rules	11
Overall literals	2, 3 per rule body
Negative literals	0, 1, 2
Minimum derived atoms	1, 3

- 120 valid synthetic combinations were generated.
- Programs use unary predicates and normal rules only, which keeps the comparison controlled while still requiring nontrivial induction.
- Each benchmark has exactly one answer set and at least one derived atom.

Why This Matters

- ASP is powerful for configuration, planning, and combinatorial reasoning, but writing the rules is often the bottleneck.
- The paper shows that strong current LLMs can already act as useful ASP modeling assistants when a solver is kept in the loop.
- Because the task is rule induction rather than text translation, success here is evidence of structured reasoning rather than prompt memorization.

Novelty / Positioning

LAMP uses the LLM to generate ASP rules directly from facts, a target answer set, and rule bounds. This differs from work that uses LLMs mainly as a front end for natural-language translation or ILASP task construction. Each candidate is checked by Clingo at inference time, without LLM fine-tuning or ILASP-style systematic hypothesis search.

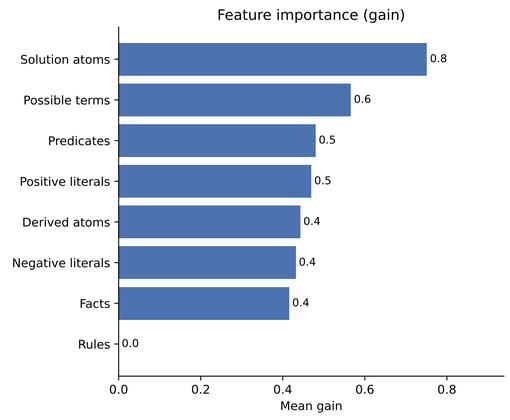
ILASP Baseline

ILASP solves the same task under the Learning from Answer Sets framework, but does so with conflict-driven symbolic search over an ASP metaprogram rather than iterative LLM generation.

- The hypothesis space is fixed by a mode bias derived from the benchmark profile.
- The target answer set is encoded as a single context-dependent positive example.
- In this study ILASP is the main symbolic baseline, run with a 420 s timeout.

Feature Analysis (RQ3)

The paper also asks which benchmark features are associated with LLM failure. The strongest models missed too few cases for a useful classifier, so the analysis focuses on *qwen3-coder:30b*.

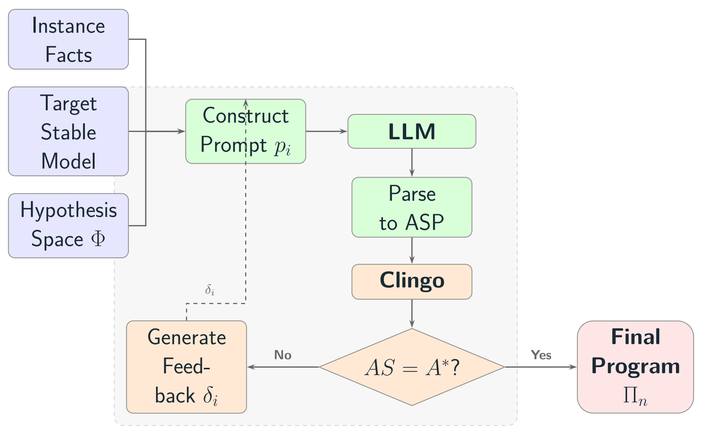


The top gain features all describe scale and vocabulary breadth: larger target answer sets, more possible terms, and more predicates make exact rule recovery harder.

XGBoost reached 61.7% cross-validation accuracy with AUC 0.619, below the 64.2% majority baseline. These are descriptive correlates, not reliable predictors yet.

LAMP Protocol

- Prompt the LLM with instance facts F, target answer set A^*, rule schema bounds, and a worked example.
- Parse the raw output into an ASP candidate program Π_i.
- Run $F \cup \Pi_i$ with Clingo.
- If the produced answer set mismatches A^*, feed the mismatch back and ask for a corrected rule set.
- Stop after success or after 3 total attempts.



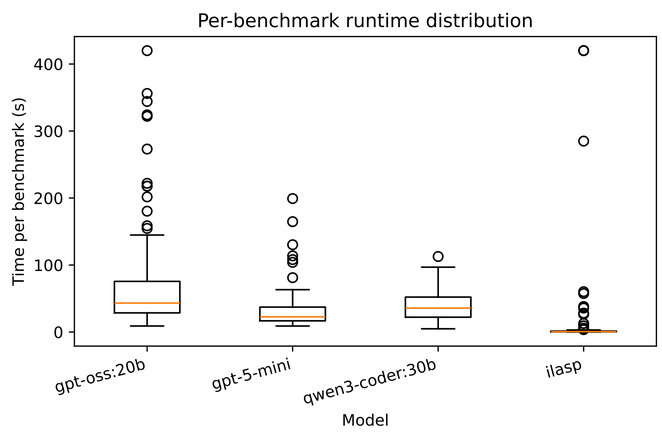
The key design choice is simple: let the LLM propose rules, but let a solver decide whether those rules are actually correct.

Accuracy Gains From Feedback

System	1st attempt	Final	Failures
gpt-5-mini	109 / 120	119 / 120	1
gpt-oss:20b	100 / 120	118 / 120	2
qwen3-coder:30b	23 / 120	77 / 120	43
ILASP	—	117 / 120	3

Feedback recovered 10 extra cases for *gpt-5-mini*, 18 for *gpt-oss:20b*, and 54 for *qwen3-coder:30b*. The loop matters, especially once the first draft is close but not exact.

Runtime And Search Behavior



Observation	Takeaway
ILASP median runtime: 0.66 s	Conflict-driven search is usually extremely fast.
ILASP failures: 3, all at 420 s	Rare search-space cliffs still matter.
Both strong LAMP models solved all 3 timeout cases	LLM-guided generation explores the space differently.

Why Hybridize?

- ILASP gives the strong symbolic baseline: usually sub-second, concise, and principled within the mode bias.
- LAMP is slower, but its token-level generation explores the space differently and solved all 3 ILASP timeout cases.
- The natural next system is not LLM versus solver; it is ILASP-style search with solver-checked LLM proposals when search stalls.

Program Quality

Mean output literal count $L(\Pi)$ for correct solutions

Original	gpt-5	gpt-oss	qwen	ILASP
22 literals	4.81	5.15	5.85	3.67
33 literals	3.49	3.82	3.84	2.71

Stratified solved programs

System	Stratified / Solved
Original benchmarks	49 / 120
gpt-5-mini	119 / 119
gpt-oss:20b	117 / 118
qwen3-coder:30b	77 / 77
ILASP	117 / 117

Both ILASP and strong LAMP models compressed the synthetic source programs heavily. Correct outputs usually needed only a handful of literals, and almost every solved program became stratified even when the original benchmark was not.

Timeout Cases

Benchmark	gpt-5	gpt-oss	qwen	ILASP
7	✓	✓	×	420 s
27	✓	✓	×	420 s
67	✓	✓	×	420 s

The three ILASP failures were all timeouts, not logical contradictions. Each timeout case had 11 negative literals, and grounded-rule counts between 20 and 27 compared with a dataset average of 13.28. That points to search-space blowup rather than impossibility of the task.

What The Results Suggest

- Strong LLMs can perform nontrivial ASP rule induction even without ASP-specific training.
- A fully local model, *gpt-oss:20b*, is already close to ILASP on this benchmark class.
- ILASP is still the faster symbolic system on average, but LAMP can succeed on some instances where exhaustive symbolic search times out.
- The generated programs are not only correct; they are often cleaner and easier to explain than the synthetic originals.

Limits And Next Steps

- The study uses synthetic benchmarks with unary predicates, normal rules, and exactly one answer set.
- Each model was run once per benchmark at temperature 1, so small score gaps should be interpreted cautiously.
- Richer targets include higher-arity ASP, choice rules, disjunction, aggregates, planning benchmarks, and ASPBench.
- The most interesting extension is a hybrid system: ILASP-style search plus an agentic LLM refinement loop.

Experimental Controls

- Same input instance for both systems: facts, target answer set, and bounded rule schema.
- LAMP uses 3 total attempts: one initial generation and up to 2 solver-feedback rounds.
- All LLMs were run once per benchmark at temperature 1.
- *gpt-5-mini* used the OpenAI API; *gpt-oss:20b* and *qwen3-coder:30b* ran locally with Ollama.
- ILASP version 4 and all LAMP runs used the same 420 s timeout.
- A LAMP output only counts as correct when Clingo returns exactly the target stable model.

Source code, benchmark data, and generated ILASP tasks: `https://github.com/dbunker/lamp`.

Research Questions Answered

RQ1	Yes: solver-in-the-loop prompting can produce correct ASP encodings from facts and a target answer set.
RQ2	LAMP nearly matches ILASP in accuracy; ILASP is faster on average, but LAMP solves the ILASP timeout cases.
RQ3	Failure correlates mostly with target size and vocabulary breadth, but the current sample is too small for reliable prediction.

Take-home

Solver-in-the-loop prompting makes ASP rule induction practical. ILASP remains the fast symbolic baseline, but LAMP provides a complementary LLM-guided search strategy: it nearly matches ILASP overall, solves the ILASP timeout cases, and often returns compact, stratified programs.